

Pseudo-code for general model calibration

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In the following, ***bold*** refers to vectorization in the variables spaces, whereas the time vectorization is implicit. When time indexing is relevant, it is explicitly reported as subscript. This pseudo-code is intended to support the user to conduct a proper model calibration, and to give a schematic representation of the ‘BIOGoALS.agri-AcoDM’ manual’s content.

Nomenclature

θ_{PSS}^*	Vector of optimal θ_{PSS} values estimated
$\theta_{P,0}$	Initial/guess literature values of θ_P
θ_{PSS}	Entries of θ_P selected by the Parameter Subset Selection (PSS) algorithm
δ^{msqr}	Vector of sensitivity magnitudes, computed as the root mean squared sensitivity indexes over the observation time points
$\epsilon_{\tilde{y}}$	Vector of relative measurements’ errors
γ	Vector of collinearity indexes
σ_a	Vector of standard deviations of the a quantity
θ_P	Uncertain process parameters
$\tilde{y}$	Measured data, already pre-processed
a(b)	Quantity a computed as a function of quantity b
COV	Covariance Matrix
e_y	Outputs’ tracking errors (residuals)
FIM	Fisher Information Matrix
SI	Matrix of sensitivity indexes
y	Vector of model’s outputs
$\mathcal{N}$	Normal Gaussian distribution

Θ	$\theta_{\mathcal{P}}$'s feasible set of values
aa	Subscripts for absolute-absolute sensitivity indexes
G	Subscripts for Global sensitivity analysis
L	Subscripts for Local sensitivity analysis
lb, ub	Subscripts for lower/upper bound
rr	Subscripts for relative-relative sensitivity indexes
MARE%	Percentage Mean Absolute Relative Error

Pseudo-code

Algorithm 1 model calibration (simplified)

- 1: Model the case study of interest in the AGRI-ACODM.MO library: create a Modelica `model`
 - 2: Convert the input data (flow rates and feedstocks' compositions) from CSV to proper TXT `CombiTimeTable` format with `GENERATE_COMBI.IPYNB`
 - 3: Integrate the Modelica `model` and plot $\mathbf{y} \rightarrow$ compute $\mathbf{e}(\boldsymbol{\theta}_{P,0}) = \tilde{\mathbf{y}} - \mathbf{y}(\boldsymbol{\theta}_{P,0})$ and model's performances (e.g. $\text{MARE}_{\%}$) for each entry of $\tilde{\mathbf{y}}$
 - 4: **if** $\text{MARE}_{\%} >$ threshold value acceptable by the user $\forall$ the entries of $\tilde{\mathbf{y}}$ **then**
 - 5: Proceed with point '9.'
 - 6: **else**
 - 7: Consider $\boldsymbol{\theta}_{PSS}^* = \boldsymbol{\theta}_{P,0}$ and move to point '14.'
 - 8: **end if**
 - 9: Perform global sensitivity analysis (GSA) with `SENSITIVITY_GLOBAL_SOBOL.IPYNB` over the most uncertain subset of $\boldsymbol{\theta}_P$ (expert-knowledge and/or literature) within $[\boldsymbol{\theta}_{P,lb}, \boldsymbol{\theta}_{P,ub}]$ i.e. $\Theta \subset \mathbb{R}_{>0}$ (expert-knowledge and/or literature) $\rightarrow \mathbf{SI}_{G,rr} \rightarrow \delta_G^{msqr} \rightarrow \rho_G \rightarrow$ identify the entries of $\boldsymbol{\theta}_P$ that globally affect $\tilde{\mathbf{y}}$ the most (highest values of δ_G^{msqr})
 - 10: Over the same subset of $\boldsymbol{\theta}_P$, perform local (around $\boldsymbol{\theta}_{P,0}$) once-at-a-time (OAT) sensitivity analysis (LSA) with `SENSITIVITY_LOCAL_OAT.IPYNB` $\rightarrow \mathbf{SI}_{L,rr} \rightarrow \delta_L^{msqr} \rightarrow \rho_L \rightarrow$ identify the entries of $\boldsymbol{\theta}_P$ that locally affect $\tilde{\mathbf{y}}$ the most (highest values of δ_L^{msqr})
 - 11: Over the same subset of $\boldsymbol{\theta}_P$, perform local FIM-based collinearity analysis with `SENSITIVITY_LOCAL_OAT.IPYNB` $\rightarrow \mathbf{FIM} \rightarrow \mathbf{COV} \rightarrow \gamma \rightarrow$ identify the combinations that are not overly collinear
 - 12: Merge the results of points '9.' to '11.' (practical identifiability analysis; Parameter Subset Selection (PSS) algorithm) $\rightarrow$ practically-identifiable $\boldsymbol{\theta}_{PSS} \subset \boldsymbol{\theta}_P$
 - 13: Estimate the $\boldsymbol{\theta}_{PSS}^*$ values that minimize the sum (over the entries of $\tilde{\mathbf{y}}$) of $\text{MARE}_{\%}$ with `PARAMETER_ESTIMATION.IPYNB`, within the Θ feasible search space.
 - 14: Repeat point '10.' and '11.' around $\boldsymbol{\theta}_{PSS}^*$, running again `SENSITIVITY_LOCAL_OAT.IPYNB`
 - 15: Considering the relative measurement noise $\epsilon_{\tilde{\mathbf{y}}}$, quantify (linear approximation) the 'residual' uncertainty associated with the estimates $\boldsymbol{\theta}_{PSS}^*$ with `PARAMETER_UNCERTAINTY_LINEAR.IPYNB` $\rightarrow \sigma_{\boldsymbol{\theta}_{PSS}^*} \rightarrow \mathcal{N}(\boldsymbol{\theta}_{PSS}^*, \sigma_{\boldsymbol{\theta}_{PSS}^*})$
 - 16: With `PARAMETER_UNCERTAINTY_LINEAR.IPYNB`, from $\mathbf{SI}_{L,aa}$ and $\mathbf{COV}$ computed at point '14.', propagate linearly the uncertainty from parameter to model's outputs $\rightarrow \sigma_{\mathbf{y}}(\boldsymbol{\theta}_{PSS}^*) \rightarrow \mathcal{N}(\mathbf{y}(\boldsymbol{\theta}_{PSS}^*), \sigma_{\mathbf{y}}(\boldsymbol{\theta}_{PSS}^*)) \rightarrow$ 95% confidence intervals (CI) of $\mathbf{y}(\boldsymbol{\theta}_{PSS}^*)$
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